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### PROFESSIONAL SUMMARY

Final year BSc Computer Engineering undergraduate with strong experience in both AI/ML and software development. Skilled in designing and deploying intelligent applications that combine modern web technologies and advanced AI technologies. Proficient in Python, FastAPI, React, TensorFlow, LangChain, n8n and Hugging Face, with practical experience in Generative AI, RAG systems, AI voice agentic systems and agentic AI architectures. Experienced in building scalable full stack applications using the NestJS and cloud platforms such as AWS and Firebase. Passionate about developing innovative, real world solutions that merge AI intelligence with robust software engineering to deliver impactful user experiences.

### EDUCATION

B.Sc. (Hons) Computer Engineering, University of Jaffna (2022 – Present)

### EXPERIENCE

AI/ML + Software Engineer Intern, Idea8 Pvt Ltd, Kottawa (2025 Nov – 2026 Apr)

### PROJECTS

#### AI Voice Agent Platform for Automated Outbound Calling ([Link](#))

**Tech:** React, n8n, Twilio, Vapi, Deepgram, ElevenLabs, Pinecone, Supabase

- Built a production ready AI voice agent system for automated outbound calling, integrating LLM powered conversational logic with real time STT (Deepgram) and TTS (ElevenLabs) pipelines.
- Implemented long term conversational memory using Pinecone vector database for semantic retrieval of historical interactions, enabling context aware, personalized responses.
- Engineered agentic workflow orchestration via n8n, coordinating telephony (Twilio), LLM inference and automated follow up actions (SMS/email) based on conversation outcomes.
- Architected a multi tenant SaaS platform with a React campaign dashboard, supporting client onboarding, contact management and real time analytics.

#### AI Powered Insurance Consultation & Meeting Intelligence Platform ([Link](#))

**Tech:** React, FastAPI, WebRTC, WebSockets, Gemini API, Pinecone, Deepgram, Supabase

- Designed an AI driven insurance consultation platform with real time video sessions (WebRTC) and a Deepgram STT pipeline for live meeting transcription and post-call storage.
- Developed a Retrieval Augmented Generation (RAG) system using Gemini and Pinecone, delivering context aware insurance recommendations grounded in internal policy knowledge bases.
- Engineered a real time AI assistant that streams LLM generated suggestions to agents during live meetings; built a meeting intelligence system with automated PDF export for compliance documentation.

#### Secure AI Medical Chatbot with Usage Control & Anti Abuse System ([LuminaAI](#))

**Tech:** React, n8n, Supabase, Google Gemini API, FingerprintJS

- Developed a secure AI medical chatbot using Google Gemini API with session memory, supporting both guest and authenticated users via a real time conversational interface.
- Engineered a multi layered usage control system with per tier prompt limits (guest vs. authenticated), persistent tracking in Supabase and usage inheritance across sessions.
- Implemented anti abuse mechanisms combining device fingerprinting (FingerprintJS) and IP based rate limiting to prevent restriction bypass.

## AI Powered Client Intake & Workflow Automation Chatbot ([BiaBot](#))

**Tech:** React, FastAPI, Supabase, OpenAI, Voice I/O, Google Cloud Run

- Built a full stack AI client intake system replacing form based onboarding with an LLM driven conversational pipeline supporting text and voice I/O (STT/TTS).
- Engineered a structured NLP intake pipeline to extract and convert natural language responses into review ready data with deterministic mission summaries.
- Integrated Monday.com API for end to end workflow automation and built audit logging for full compliance and traceability of client submissions.

## Scalable HR Management Platform for Idea8 Pvt Ltd

**Tech:** NestJS (TypeScript), React, PostgreSQL, Prisma, WebSockets

- Designed a production grade HR platform with multi role access control (Admin, HR, Employee), JWT authentication and a modular NestJS backend with Prisma ORM.
- Implemented a real time notification system using WebSockets for instant HR event alerts and scheduled cron jobs for automated email notifications via Nodemailer.
- Built workflow driven leave and WFH approval pipelines with full auditability; containerized with Docker for consistent AWS deployment.

## Multimodal AI System for Eye Disease Diagnosis (Research)

**Tech:** CNN (InceptionV3), BERT, ViT, RAG, FAISS, Google Gemini, PyTorch, TensorFlow, Streamlit

- Engineered a multimodal diagnostic system with three inference pipelines: CNN (InceptionV3) for retinal/OCT image classification, BERT for clinical NLP reasoning and a joint multimodal fusion model combining image and text embeddings.
- Built a RAG pipeline using FAISS and Google Gemini for medical knowledge grounding and context aware clinical question answering.
- Implemented a chat driven diagnostic assistant with LLM generated explanations for predicted conditions, improving interpretability for non expert users.

## Multimodal Early Warning System for Business Sales Risk Prediction (Ongoing)

**Tech:** Scikit-learn, TensorFlow, OpenCV, NLP, Computer Vision, Pandas, Feature Engineering

- Developed a multimodal ML system combining NLP sentiment analysis, CNN based image classification (product defect detection) and financial risk modeling to predict sales performance.
- Engineered temporal behavioral features (engagement decay, sentiment ratios) and a multi domain feature fusion framework integrating behavioral, visual and financial signals.

## SKILLS AND ABILITIES

- Programming Languages: Python, Java, C++
- SE Stack: MERN, NestJS
- ML and DL Frameworks: Scikit-learn, TensorFlow, Keras, Random Forest, Regression Models
- Generative AI & NLP: LLMs, LangChain, Transformers, RAG, Prompt Engineering, Gemini, Groq, GPT models
- Agentic AI & Automation: Agents, LangGraph, n8n
- Web & API Development: FastAPI, Streamlit, Flask, React, REST APIs
- Databases & Vector Databases: PostgreSQL, MongoDB, Supabase, MySQL, ChromaDB, Pinecone
- DevOps & Tools: Google Colab, Hugging Face, Vercel, Railway, LLMops, Docker, AWS EC2, Vapi, ElevenLabs, Google cloud, Cloudways

## REFERENCES

- Dr. Pratheeba Jeyanathan, Head of the department, Dept. of Computer Engineering Faculty of Engineering, University of Jaffna, [pratheeba@eng.jfn.ac.lk](mailto:pratheeba@eng.jfn.ac.lk), +94778448679
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